

Piping Material & Service Cost Estimate Report

Consolidated management review with material-cost and direct-service-cost backup

Draft for review

PROJECT / JOB DESCRIPTION

Demo

PROJECT NO. / MOC NO. / JOB NO.

Demo No-0001-2026

DESIGN TEMPERATURE (°C)

Not specified

TOTAL BASE PROJECT COST

Rs 129.60 Lakhs

● Material and direct service estimate ready for review

Material base plus direct service cost. Taxes, freight, escalation, contingency and commercial additions are excluded.

Report generated: 16 Jul 2026, 05:17 pm | Year basis: 2026 | Consolidated budgetary estimate

PROJECT COST COMPOSITION



TOTAL BASE PROJECT COST

Rs 129.60 Lakhs

Part A material base + Part B direct service.

MATERIAL RISK RESERVE

Rs 35.51 Lakhs

Restricted to Part A material estimate.

TOTAL APPROVAL ENVELOPE

Rs 165.12 Lakhs

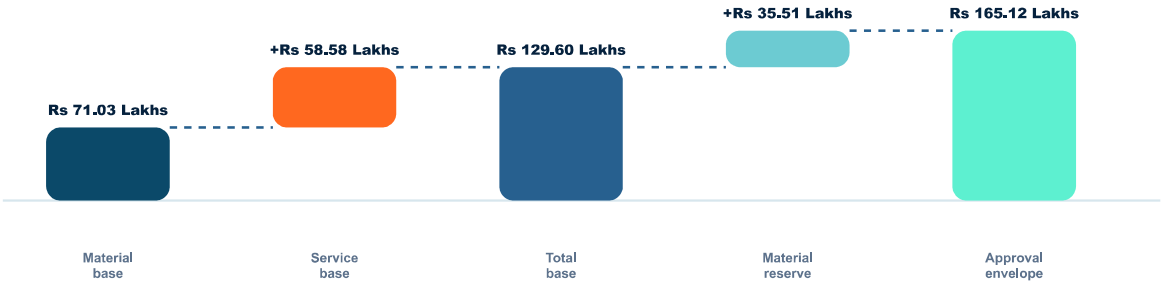
Part A P90 material + Part B service base.

ESTIMATE READINESS

37 priced / 1 review

Pending items need a management decision.

Project Approval Bridge



Material base + service base = total base; add the Part A material reserve for the approval envelope.

APPROVAL REQUEST

Rs 165.12 Lakhs

Approve the consolidated budget envelope.

RESERVE DISCIPLINE

27.4%

Apply reserve to Part A material cost only.

CRITICAL VALIDATION

Pipe Group

Validate the highest material cost driver.

PENDING CONDITION

1 review item(s)

Price or dispose of review-required items.

Part A - Piping Material Cost

Material estimate with factor backup, explicit reserve and P90 approval envelope.

Material reserve retained

Piping Material Cost Summary

BASE ESTIMATE

Rs 71.03 Lakhs

Normal case for planning and comparison.

P90 ESTIMATE

Rs 106.54 Lakhs

Conservative material envelope for approval.

RISK RESERVE

Rs 35.51 Lakhs

Difference between P90 and base estimate.

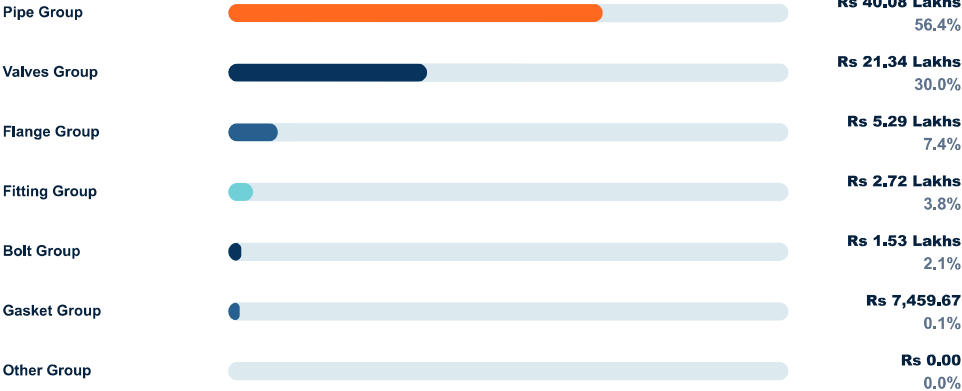
MAIN COST DRIVER

Pipe Group

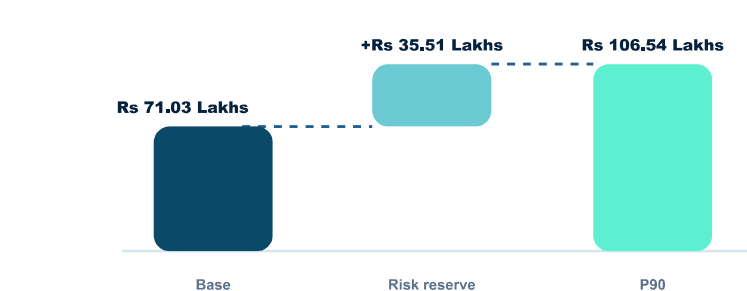
56.4% share of material base.

Group-wise Cost

Total BOM items = 22



Estimate Bridge



Visual bridge: base estimate plus risk reserve equals the P90 budget envelope.

Management focus

Check the highest-value group first before approval.

Risk focus

Keep the reserve visible so exposure is clear.

Data quality

Some items still need a confirmed price or management decision.

Component Cost Summary

Group-level material cost, reserve and action status for management review.

1 review item(s)

GROUP	ITEMS	BASE COST	P90 COST	RESERVE	SHARE	STATUS	REQUIRED ACTION
Pipe Group	3	Rs 40.08 Lakhs	Rs 60.13 Lakhs	Rs 20.04 Lakhs	56.4%	Priority	Validate rate source, class and factor
Valves Group	2	Rs 21.34 Lakhs	Rs 32.01 Lakhs	Rs 10.67 Lakhs	30.0%	Checked	Routine check
Flange Group	5	Rs 5.29 Lakhs	Rs 7.94 Lakhs	Rs 2.65 Lakhs	7.4%	Checked	Routine check
Fitting Group	6	Rs 2.72 Lakhs	Rs 4.07 Lakhs	Rs 1.36 Lakhs	3.8%	Checked	Routine check
Bolt Group	3	Rs 1.53 Lakhs	Rs 2.29 Lakhs	Rs 76,257.16	2.1%	Checked	Routine check
Gasket Group	2	Rs 7,459.67	Rs 11,189.50	Rs 3,729.83	0.1%	Checked	Routine check
Other Group	1	Rs 0.00	Rs 0.00	Rs 0.00	0.0%	Pending	1 (One) item needs review to calculate the group total.

Part A - Material Cost Detail

Line-level calculation backup by piping component group.

Audit backup

1 Pipe Group

3 item(s) | Normal Rs 40.08 Lakhs | P90 Rs 60.13 Lakhs

ITEM	SIZE	RATING / THK	MATERIAL	MATERIAL CATEGORY	QTY	UOM	FACTOR	UNIT RS	NORMAL TOTAL	P90 TOTAL
PIPE SEAMLESS	12"	22 mm	ASTM A106 GR.B	Carbon Steel	240	M	1.00	Rs 16,658.13	Rs 39.98 Lakhs	Rs 59.97 Lakhs
PIPE SEAMLESS	2"	XS	ASTM A106 GR.B	Carbon Steel	12	M	1.00	Rs 760.88	Rs 9,130.51	Rs 13,695.76
PIPE SEAMLESS	3/4"	XS	ASTM A106 GR.B	Carbon Steel	6	M	1.00	Rs 223.49	Rs 1,340.95	Rs 2,011.43

2 Fitting Group

6 item(s) | Normal Rs 2.72 Lakhs | P90 Rs 4.07 Lakhs

ITEM	SIZE	RATING / THK	MATERIAL	MATERIAL CATEGORY	QTY	UOM	FACTOR	UNIT RS	NORMAL TOTAL	P90 TOTAL
ELBOW 90D	12"	S80	ASTM A234 GR.WPB	Carbon Steel	8	Nos.	2.50	Rs 14,215.08	Rs 1.14 Lakhs	Rs 1.71 Lakhs
ELBOW 45D	12"	S80	ASTM A234 GR.WPB	Carbon Steel	6	Nos.	2.50	Rs 8,529.05	Rs 51,174.28	Rs 76,761.42
TEE EQUAL	12"	S80	ASTM A234 GR.WPB	Carbon Steel	6	Nos.	3.86	Rs 17,258.29	Rs 1.04 Lakhs	Rs 1.55 Lakhs
ELBOW 90D	2"	XS	ASTM A234 GR.WPB	Carbon Steel	5	Nos.	5.00	Rs 300.26	Rs 1,501.32	Rs 2,251.98
ELBOW 90D	3/4"	3000#	ASTM A105	Carbon Steel	4	Nos.	0.55	Rs 161.71	Rs 646.83	Rs 970.24
COUPLING HALF	3/4"	3000#	ASTM A105	Carbon Steel	8	Nos.	0.45	Rs 132.31	Rs 1,058.45	Rs 1,587.67

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Flange Group

5 item(s) | Normal Rs 5.29 Lakhs | P90 Rs 7.94 Lakhs

ITEM	SIZE	RATING / THK	MATERIAL	MATERIAL CATEGORY	QTY	UOM	FACTOR	UNIT RS	NORMAL TOTAL	P90 TOTAL
FLANGE WN	12"	600#	ASTM A105	Carbon Steel	14	Nos.	3.70	Rs 21,532.15	Rs 3.01 Lakhs	Rs 4.52 Lakhs
BLIND FLANGE	12"	600#	ASTM A105	Carbon Steel	4	Nos.	2.90	Rs 20,317.40	Rs 81,269.60	Rs 1.22 Lakhs
SPACER & BLIND	12"	600#	ASTM A105	Carbon Steel	7	Nos.	2.90	Rs 20,317.40	Rs 1.42 Lakhs	Rs 2.13 Lakhs
FLANGE SW	3/4"	600#	ASTM A105	Carbon Steel	8	Nos.	3.70	Rs 303.12	Rs 2,424.98	Rs 3,637.47
BLIND FLANGE	3/4"	600#	ASTM A105	Carbon Steel	8	Nos.	2.90	Rs 209.73	Rs 1,677.82	Rs 2,516.74

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Valves Group

2 item(s) | Normal Rs 21.34 Lakhs | P90 Rs 32.01 Lakhs

ITEM	SIZE	RATING / THK	MATERIAL	MATERIAL CATEGORY	QTY	UOM	FACTOR	UNIT RS	NORMAL TOTAL	P90 TOTAL
BALL VALVE	12"	600#	BODY-ASTM A 105/A 216GR.WCB,TRIM-SEAT-RPTFE	Carbon Steel	7	Nos.	3.70	Rs 2.91 Lakhs	Rs 20.40 Lakhs	Rs 30.60 Lakhs
BALL VALVE	3/4"	600#	BODY-ASTM A 105/A 216GR.WCB,TRIM-SEAT-RPTFE	Carbon Steel	8	Nos.	4.20	Rs 11,741.16	Rs 93,929.31	Rs 1.41 Lakhs

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Bolt Group

3 item(s) | Normal Rs 1.53 Lakhs | P90 Rs 2.29 Lakhs

ITEM	SIZE	RATING / THK	MATERIAL	MATERIAL CATEGORY	QTY	UOM	SET WT (KG)	FACTOR	UNIT RS	NORMAL TOTAL	P90 TOTAL
STUDS & NUTS	M33x240	-	A193 GR. B7 & A194 GR.2H	Unclassified	280	Nos.	2.338	2.50	Rs 330.28	Rs 92,478.66	Rs 1.39 Lakhs
STUDS & NUTS	M33x350	-	A193 GR. B7 & A194 GR.2H	Unclassified	140	Nos.	2.979	2.50	Rs 420.72	Rs 58,901.26	Rs 88,351.89
STUDS & NUTS	M16x100	-	A193 GR. B7 & A194 GR.2H	Unclassified	32	Nos.	0.251	2.50	Rs 35.45	Rs 1,134.40	Rs 1,701.60

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Gasket Group

2 item(s) | Normal Rs 7,459.67 | P90 Rs 11,189.50

ITEM	SIZE	RATING / THK	MATERIAL	MATERIAL CATEGORY	QTY	UOM	FACTOR	UNIT RS	NORMAL TOTAL	P90 TOTAL
GASKET SPIRAL	12"	5MM	SS304 SPR WND + CA FIL	Unclassified	30	Nos.	0.06	Rs 239.95	Rs 7,198.42	Rs 10,797.64
GASKET SPIRAL	3/4"	5MM	SS304 SPR WND + CA FIL	Unclassified	16	Nos.	0.06	Rs 16.33	Rs 261.24	Rs 391.86

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Other Group

1 item(s) | Normal Rs 0.00 | P90 Rs 0.00

ITEM	SIZE	RATING / THK	MATERIAL	MATERIAL CATEGORY	QTY	UOM	FACTOR	UNIT RS	NORMAL TOTAL	P90 TOTAL
PRESSURE GAUGE WITH ACCESSORIES	0-60KG/CM2	AS PER MNF. STD.	-	Unclassified	1	Nos.	0.00	Review	Review	Review

Part B - Piping Service Cost

Direct service estimate using schedule-of-rates and derived quantity logic.

No service reserve applied

Piping Service Cost Summary

No Estimate Bridge

DIRECT SERVICE COST

Rs 58.58 Lakhs

Taxes, escalation and contingency excluded.

CORE PIPING INSTALLATION

Rs 42.55 Lakhs

72.6% of service cost.

ASSOCIATED WORKS

Rs 9.20 Lakhs

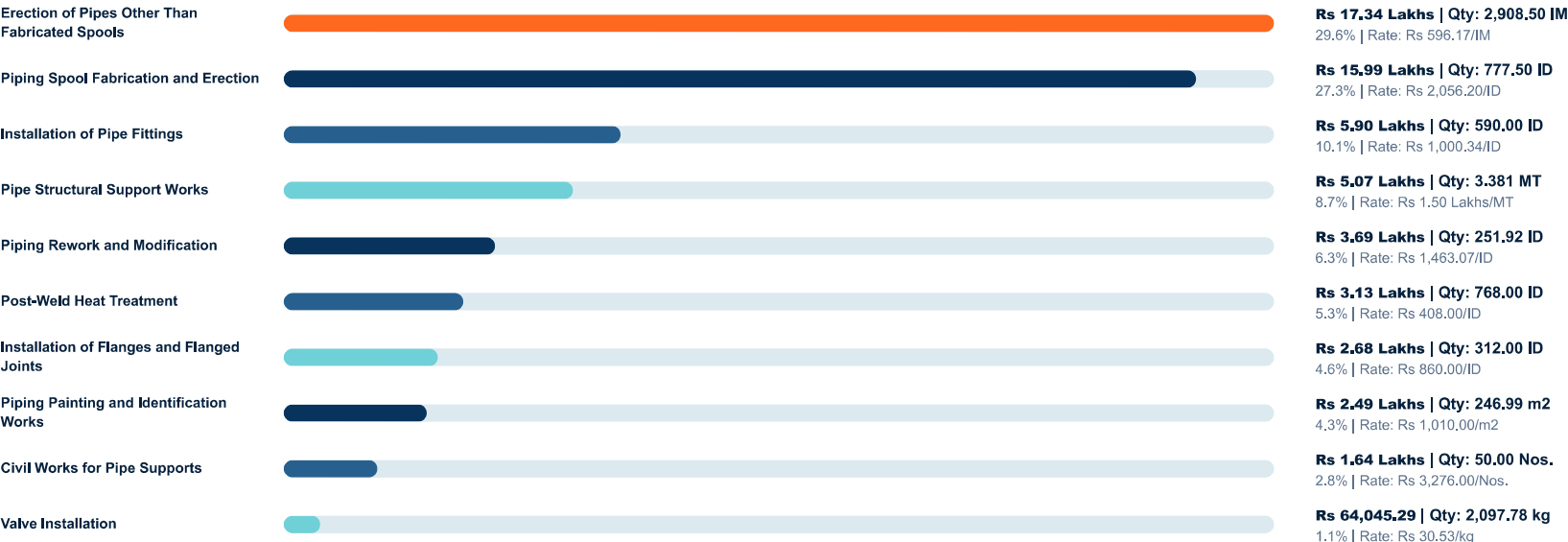
15.7% of service cost.

MAIN COST DRIVER

Erection of Pipes Other Than Fabricated Spools

Rs 17.34 Lakhs direct cost.

Service Item Cost Distribution



Items are ordered from highest to lowest direct service cost.

Service Basis, Quantities & Cost Build-up

Key work volumes and preliminary direct-service cost allowances.

Direct SOR cost only

① Piping Spool Fabrication and Erection

Qty: 777.50 ID

Cost: Rs 15.99 Lakhs
Rate: Rs 2,056.20/ID

② Installation of Pipe Fittings

Qty: 590.00 ID

Cost: Rs 5.90 Lakhs
Rate: Rs 1,000.34/ID

③ Installation of Flanges and Flanged Joints

Qty: 312.00 ID

Cost: Rs 2.68 Lakhs
Rate: Rs 860.00/ID

④ Erection of Pipes Other Than Fabricated Spools

Qty: 2,908.50 IM

Cost: Rs 17.34 Lakhs
Rate: Rs 596.17/IM

⑤ Valve Installation

Qty: 2,097.78 kg

Cost: Rs 64,045.29
Rate: Rs 30.53/kg

⑥ Piping Rework and Modification

Qty: 251.92 ID

Cost: Rs 3.69 Lakhs
Rate: Rs 1,463.07/ID

⑦ Pipe Structural Support Works

Qty: 3.381 MT

Cost: Rs 5.07 Lakhs
Rate: Rs 1.50 Lakhs/MT

⑧ Civil Works for Pipe Supports

Qty: 50.00 Nos.

Cost: Rs 1.64 Lakhs
Rate: Rs 3,276.00/Nos.

⑨ Piping Painting and Identification Works

Qty: 246.99 m2

Cost: Rs 2.49 Lakhs
Rate: Rs 1,010.00/m2

⑩ Piping Insulation Works

Qty: No eligible quantity

Cost: Rs 0.00
Rate: Rate requires review

⑪ Post-Weld Heat Treatment

Qty: 768.00 ID

Cost: Rs 3.13 Lakhs
Rate: Rs 408.00/ID

SERVICE BASIS

Above Ground / Non-IBR

ELIGIBLE SERVICE ROWS

16

NEED REVIEW

0

DIRECT SERVICE COST

Rs 58.58 Lakhs

Service Cost Category Summary

Direct SOR cost only

CATEGORY	SOURCE ROWS	DIRECT COST	SHARE	STATUS	MANAGEMENT REVIEW
Core piping installation	Rows 1-5	Rs 42.55 Lakhs	72.6%	Ready	Focus on fabrication, erection and rate source
Associated works	Rows 7-10	Rs 9.20 Lakhs	15.7%	Ready	Support, civil, painting and insulation
Modification & special processes	Rows 6 and 11	Rs 6.82 Lakhs	11.6%	Ready	Rework and PWHT
Total service cost	Rows 1-11	Rs 58.58 Lakhs	100.0%	Ready	Taxes, escalation and contingency excluded

Do not add the material reserve percentage automatically to Part B. Any future service contingency should be based on approved quantity, productivity, site-condition or schedule uncertainty.

Part B - Service Schedule-of-Rates Detail

Work-package quantity, rate, cost basis and review status | 11 priced work packages | 0 need review

Direct service cost: Rs 58.58 Lakhs

SL.	SHORT DESCRIPTION	JOB DESCRIPTION / ACTIVITIES	UNIT	QUANTITY	RATE	AMOUNT	RATE SOURCE / NOTE	STATUS
1	Piping Spool Fabrication and Erection	Fabrication, cutting, edge preparation, fit-up, welding, handling, transportation, erection, alignment, NDT, pressure testing or hydrostatic testing, flushing and reinstatement, as applicable.	ID	777.50	Rs 2,056.20	Rs 15.99 Lakhs	Approved material-category ID rate library	Ready
2	Installation of Pipe Fittings	Cutting, fit-up, welding and installation of elbows, tees, reducers, caps, couplings and other piping fittings, including alignment, NDT and pressure testing or hydrostatic testing, as applicable.	ID	590.00	Rs 1,000.34	Rs 5.90 Lakhs	Matching-pipe ID rate x fitting connection-end proxy	Ready
3	Installation of Flanges and Flanged Joints	Fabrication, fit-up, welding and erection of flanges, including alignment, NDT, pressure testing, gasket installation, bolting, tightening and final box-up of flanged joints.	ID	312.00	Rs 860.00	Rs 2.68 Lakhs	Matching-pipe ID rate x 1 joint per flange	Ready
4	Erection of Pipes Other Than Fabricated Spools	Handling, transportation, lifting, positioning, erection, alignment and securing of straight pipes or piping components supplied as loose items, excluding shop-fabricated or field-fabricated spools.	IM	2,908.50	Rs 596.17	Rs 17.34 Lakhs	Approved material-category IM rate library	Ready
5	Valve Installation	Transportation, handling, lifting, positioning, erection and assembly of valves, including gasket installation, bolting, alignment, PMI, functional testing, pressure testing and final box-up, as applicable.	kg	2,097.78	Rs 30.53	Rs 64,045.29	IOCL-ME-SOR: RTJ / non-RTJ valve kg rate	Ready
6	Piping Rework and Modification	Dismantling, cutting, removal, re-routing, modification and reinstallation of existing piping, fittings, flanges, valves and supports, including edge preparation, fit-up, welding, NDT, PWHT, pressure testing, touch-up painting and reinstatement, as applicable.	ID	251.92	Rs 1,463.07	Rs 3.69 Lakhs	15.00% of eligible project welding ID and matched welding cost	Ready
7	Pipe Structural Support Works	Supply, fabrication, cutting, drilling, welding, surface preparation, priming, transportation, erection, alignment and installation of structural steel pipe supports, including clamps, guides, shoes, brackets, stools and associated hardware.	MT	3.381	Rs 1.50 Lakhs	Rs 5.07 Lakhs	NRL LPP/WO adopted rate: Rs 1,50,000/MT	Ready
8	Civil Works for Pipe Supports	Excavation, PCC, RCC, reinforcement, formwork, foundation construction, installation of anchor bolts or embedded plates, non-shrink grouting, curing, backfilling, compaction and disposal of surplus excavated material.	Nos.	50.00	Rs 3,276.00	Rs 1.64 Lakhs	Inside Unit Battery Limit (30%) x 300 mm above-ground, size-based civil cost per support: 3 IN = Rs 6,000; 20 IN = Rs 17,000; 48 IN = Rs 26,000; linear interpolation and upward rounding to Rs 500.	Ready
9	Piping Painting and Identification Works	Supply of all materials and execution of surface preparation, abrasive blasting, primer application, intermediate and finish coats, touch-up painting, colour coding, flow-direction marking and line identification, as per the approved painting specification.	m2	246.99	Rs 1,010.00	Rs 2.49 Lakhs	65C default uninsulated-piping P50 rate 325 microns	Ready
10	Piping Insulation Works	Supply and installation of insulation material, cladding, bands, fasteners, sealants and accessories, including surface preparation, joint sealing, weatherproofing and completion of insulation at fittings, valves, flanges and supports.	m2	-	-	-	Insulation rate requires Design Temperature	Not applicable
11	Post-Weld Heat Treatment	Carrying out PWHT of weld joints, including heating arrangement, thermocouple installation, controlled heating, soaking and cooling, temperature monitoring and chart recording, hardness testing and submission of PWHT reports, as applicable.	ID	768.00	Rs 408.00	Rs 3.13 Lakhs	NRL PWHT rule and material-category ID rate	Ready

Direct Service Cost Total

Rs 58.58 Lakhs

Taxes, escalation and contingency excluded

Part B - Service Schedule-of-Rates Detail

Work-package quantity, rate, cost basis and review status.

Schedule-of-rates backup

1 Pipe Service

3 item(s) | Direct service Rs 33.33 Lakhs | 0 review

Carbon Steel

COMPONENT	SIZE	THK MM	BOM MATERIAL	LENGTH M	EST. JOINTS	ERECTION IM	WELDING ID	ERECTION RATE	WELDING RATE	ERECTION COST	WELDING COST	DIRECT SERVICE COST	STATUS / RATE SOURCE
PIPE SEAMLESS	12 IN	22.00	ASTM A106 GR.B	240.00	64	2,880.00	768.00	Rs 599.00	Rs 2,071.00	Rs 17.25 Lakhs	Rs 15.91 Lakhs	Rs 33.16 Lakhs	Carbon Steel Derived in uploaded document Derived in uploaded document
PIPE SEAMLESS	2 IN	5.54	ASTM A106 GR.B	12.00	4	24.00	8.00	Rs 310.00	Rs 860.00	Rs 7,440.00	Rs 6,880.00	Rs 14,320.00	Carbon Steel BPCL-Kochi BPCL-Kochi
PIPE SEAMLESS	3/4 IN	3.91	ASTM A106 GR.B	6.00	2	4.50	1.50	Rs 310.00	Rs 860.00	Rs 1,395.00	Rs 1,290.00	Rs 2,685.00	Carbon Steel BPCL-Kochi BPCL-Kochi

2 Fitting Service

6 item(s) | Direct service Rs 5.90 Lakhs | 0 review

Carbon Steel

COMPONENT	SIZE	THK MM	BOM MATERIAL	QTY.	EST. JOINTS	WELDING ID	WELDING RATE	WELDING COST	DIRECT SERVICE COST	STATUS / RATE SOURCE
ELBOW 90D	12 IN	17.45	ASTM A234 GR.WPB	8.00	16	192.00	Rs 1,010.00	Rs 1.94 Lakhs	Rs 1.94 Lakhs	Carbon Steel BPCL-Kochi
ELBOW 45D	12 IN	17.45	ASTM A234 GR.WPB	6.00	12	144.00	Rs 1,010.00	Rs 1.45 Lakhs	Rs 1.45 Lakhs	Carbon Steel BPCL-Kochi
TEE EQUAL	12 IN	17.45	ASTM A234 GR.WPB	6.00	18	216.00	Rs 1,010.00	Rs 2.18 Lakhs	Rs 2.18 Lakhs	Carbon Steel BPCL-Kochi
ELBOW 90D	2 IN	5.54	ASTM A234 GR.WPB	5.00	10	20.00	Rs 860.00	Rs 17,200.00	Rs 17,200.00	Carbon Steel BPCL-Kochi
ELBOW 90D	3/4 IN	5.54	ASTM A105	4.00	8	6.00	Rs 860.00	Rs 5,160.00	Rs 5,160.00	Carbon Steel BPCL-Kochi
COUPLING HALF	3/4 IN	5.54	ASTM A105	8.00	16	12.00	Rs 860.00	Rs 10,320.00	Rs 10,320.00	Carbon Steel BPCL-Kochi

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Flange Service

5 item(s) | Direct service Rs 2.68 Lakhs | 0 review

Carbon Steel

COMPONENT	SIZE	THK MM	BOM MATERIAL	QTY.	EST. JOINTS	WELDING ID	WELDING RATE	WELDING COST	DIRECT SERVICE COST	STATUS / RATE SOURCE
FLANGE WN	12 IN	9.53	ASTM A105	14.00	14	168.00	Rs 860.00	Rs 1.44 Lakhs	Rs 1.44 Lakhs	Carbon Steel BPCL-Kochi
BLIND FLANGE	12 IN	9.53	ASTM A105	4.00	4	48.00	Rs 860.00	Rs 41,280.00	Rs 41,280.00	Carbon Steel BPCL-Kochi
SPACER & BLIND	12 IN	9.53	ASTM A105	7.00	7	84.00	Rs 860.00	Rs 72,240.00	Rs 72,240.00	Carbon Steel BPCL-Kochi
FLANGE SW	3/4 IN	2.87	ASTM A105	8.00	8	6.00	Rs 860.00	Rs 5,160.00	Rs 5,160.00	Carbon Steel BPCL-Kochi
BLIND FLANGE	3/4 IN	2.87	ASTM A105	8.00	8	6.00	Rs 860.00	Rs 5,160.00	Rs 5,160.00	Carbon Steel BPCL-Kochi

4

Valve Service

2 item(s) | Direct service Rs 64,045.29 | 0 review

Carbon Steel

COMPONENT	SIZE	BOM MATERIAL	QTY.	VALVE WT KG	SERVICE RATE	VALVE SERVICE COST	DIRECT SERVICE COST	STATUS / RATE SOURCE
BALL VALVE	12"	BODY-ASTM A 105/A 216GR.WCB,TRIM-SEAT-RPTFE	7.00	288.00	Rs 30.53/kg	Rs 61,548.48	Rs 61,548.48	IOCL-ME-SOR Valve other than RTJ Rs 30.53/kg
BALL VALVE	3/4"	BODY-ASTM A 105/A 216GR.WCB,TRIM-SEAT-RPTFE	8.00	10.22	Rs 30.53/kg	Rs 2,496.81	Rs 2,496.81	IOCL-ME-SOR Valve other than RTJ Rs 30.53/kg

Calculation Methodology & Audit Trail

Key pricing basis, derived quantities, rate sources and exclusions.

Part A - Material Cost Methodology

- **Component pricing:** each component follows its approved group method and quantity basis.
- **Pipe weight:** $W = 0.0246615 \times (OD - t) \times t$, where OD is actual outside diameter in mm and t is wall thickness in mm. Total pipe weight = kg/m x length.
- **90 degree LR elbows:** developed length = $2.356 \times \text{actual OD}$; elbow weight = pipe kg/m x developed length. CS factor bands are 2 IN = 5.00; above 2 to 4 IN = 3.50; above 4 to 20 IN = 2.50; above 20 to 26 IN = 2.80; above 26 to 48 IN = 3.20. Unit Rs = elbow weight x raw material Rs/kg x factor x material multiplier.
- **45 degree elbows:** use 50% of matching 90 degree elbow physical weight. Unit price = 65% of matching 90 degree elbow price for NPS 2 to 6 IN, and 60% for NPS 8 to 48 IN.
- **Equal tees:** developed length = $2C + M - 0.5 \times OD$; tee weight = pipe kg/m x developed length. PO-derived factors are NPS 2 to 4 IN = 4.41, NPS 6 to 20 IN = 3.86, and NPS 24 to 48 IN = 3.55.
- **Reducing tees:** approved unequal-tee C and M dimensions define the run and branch weight portions; the same PO-derived Equal Tee factor is used for the run size. Where dimensions cannot produce a price, the run-size Equal Tee price is used and labelled Equal tee fallback.
- **Flanges:** WN, SO and Blind flange cost = JSON flange weight x raw material Rs/kg x P50 base multiplier x quantity.
- **Metric stud sets:** cost = calculated stud-and-two-heavy-hex-nut mass x raw material Rs/kg x commercial raw-to-finished factor 2.50 x quantity. ASME B18.2.4.6M heavy-hex nut dimensions are used.
- **Other components:** non-pipe components continue to use their approved component pricing methods.
- **Excluded from stud-set mass:** washers, coatings, chamfers, thread tolerances and manufacturing variation.
- **Commercial scope:** this is an indicative material and direct-service estimate. Taxes, freight, escalation, contingency, packing, wastage, contractor margin and commercial terms are excluded unless separately stated.

Part B - Service Cost Methodology

- **Rate library:** Carbon Steel uses CS rates, Austenitic Stainless Steel uses SS rates, and Alloy Steel uses AS rates. Unclassified rows use CS as a visible fallback; other unsupported categories remain Review.
- **Pipe erection and welding:** each pipe row is one run. Base joints = ceiling(pipe length / 6 m); estimated joints = ceiling(base joints x 1.60). Erection quantity = NPS x length in IM; welding quantity = estimated joints x NPS in ID.
- **PWHT:** applied only when BOM Piping Class / PMS Class, material and wall thickness match `nr1_pwht_rules`. Eligible rates are CS/LTCS Rs 408/ID, P11/P12/P22 Rs 559/ID, P5/P9 Rs 613/ID, and P91/P92 Rs 715/ID. Austenitic Stainless Steel uses the Rs 559/ID alloy-steel rate.
- **Pipe supports:** structural steel per support = matching pipe kg/m x 0.50 m support height. Total structural steel MT = per-support steel x support count, using suggested maximum spans: 1 IN = 2 m, 2 IN = 3 m, 3-4 IN = 4 m, and 6-48 IN = 6 m; then x Rs 1,50,000/MT. Civil support cost for a 300 mm above-ground support = individual support count x size-based civil cost per support: 3 IN = Rs 6,000, 20 IN = Rs 17,000 and 48 IN = Rs 26,000, linearly interpolated and rounded upward to the next Rs 500. Inside Unit Battery Limit uses 30% of this civil cost; Outside Unit Battery Limit uses 100%.
- **Insulation and painting:** insulation = $\pi \times OD (m) \times \text{length} (m) \times \text{provisional P50 Rs/m}^2 \text{ at Design Temperature}$. Painting uses the same surface area: Rs 1,010/m² at the 65C default; under-insulation rates are Rs 885/m² from 65C to 200C, interpolate to Rs 2,220/m² from 200C to 300C, then remain Rs 2,220/m² to 500C.
- **Valve service:** calculated valve material-cost weight x BOM quantity x IOCL-ME-SOR rate.
- **Excluded:** commissioning, taxes, escalation and contingency.

This report is intended for budgetary planning and management review. Final commercial value shall be validated through latest approved rate source, vendor quotation, PMS/specification, quantity take-off, taxes, freight, inspection, delivery basis, site constraints and commercial conditions.